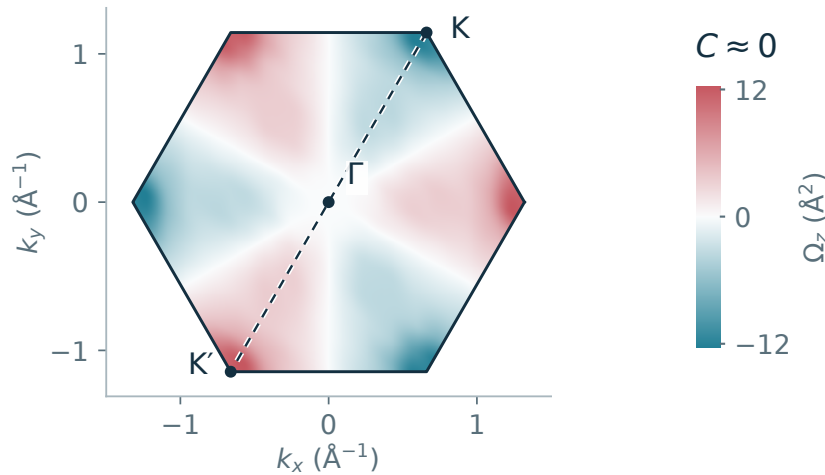


2D topology, transport, optics & wavefunctions - from VASP outputs

01 Berry curvature · map & path

Methods: Fukui-Hatsugai-Suzuki, JPSJ 74, 1674 (2005) · Kubo (1957) · Wang et al., PRB 74, 195118 (2006)

FHS · finite-plaquette map

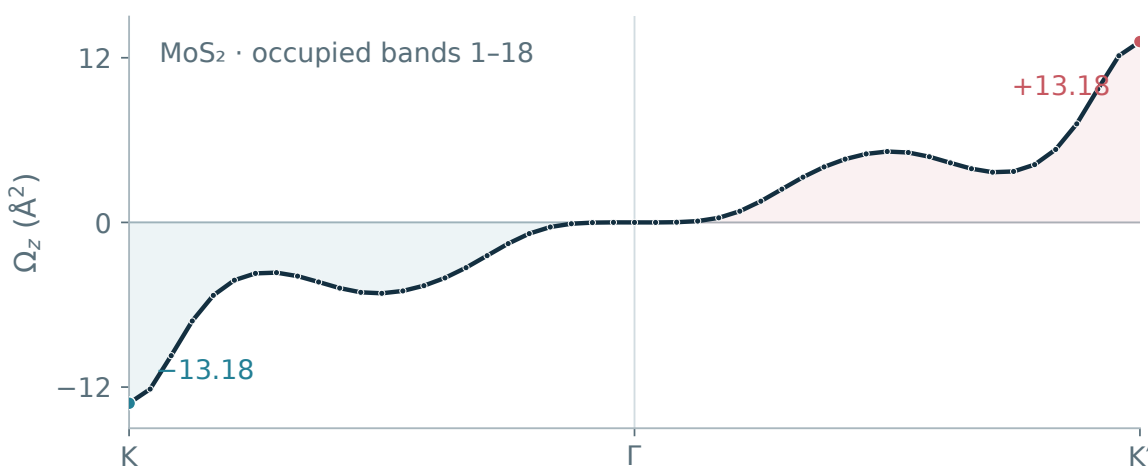


MoS₂ · occupied bands 1-18 · FHS
12 × 12 mesh; periodic cubic display only

$$\Phi_p = -\text{Arg} \prod_{l \in \partial p} U_l, \quad \tilde{\Omega}_p = \Phi_p / \Delta S_k$$

Fukui-Hatsugai-Suzuki (FHS): oriented overlap links; Φ_p is the Berry flux.

Kubo · independently evaluated k-path



49 actual path points · independent of map
WAVECAR bare-momentum approximation

$$\Omega_{xy}^V = -2\hbar^2 \text{Im} \sum_{n \in V, m \notin V} \frac{v_{nm}^x v_{mn}^y}{(E_n - E_m)^2}$$

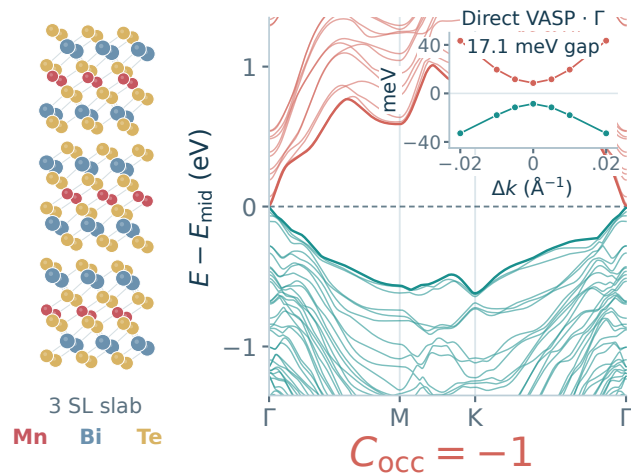
Kubo route: V = bands 1-18; empty 19-26; bare velocity $v = p / m_e$.

02 Chern & spin topology

Methods / material: Fukui et al. (2005) · Prodan (2009) · Marzari-Vanderbilt (1997) · Souza et al. (2002) · Murakami (2006)

MnBi₂Te₄ · charge Chern

VASP-derived Wannier bands



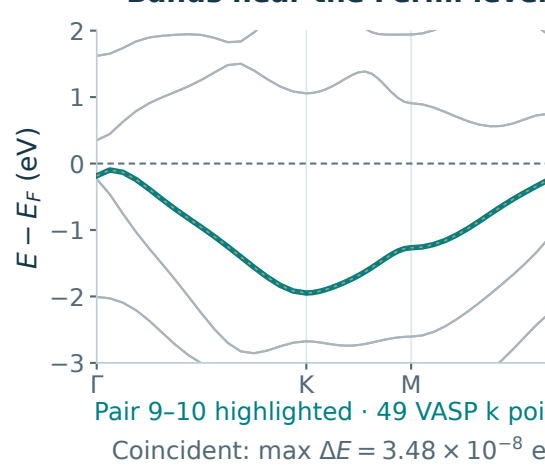
Full-BZ FHS · native WAVECAR
Occupied VASP bands 1-123

$$H^W(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} H^W(\mathbf{R}), \quad C_{\text{occ}} = \frac{1}{2\pi} \sum_p \Phi_p^{\text{occ}}$$

MBT bands: Wannier Hamiltonian; Cocc: independent native occupied-space FHS.

Bi bilayer · degenerate pair in full band context

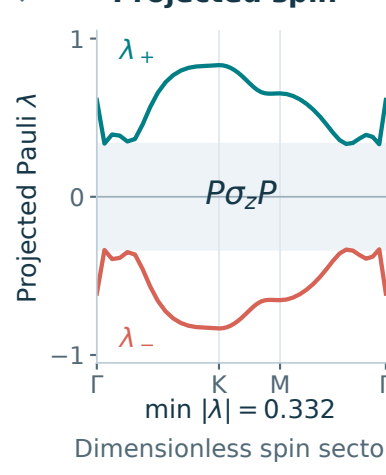
Bands near the Fermi level



Full occupied 1-10: C_{spin} = 1

Independent FHS results · native 12×12 full BZ · WAVECAR pseudo-Gram (no PAW augmentation)

Projected spin



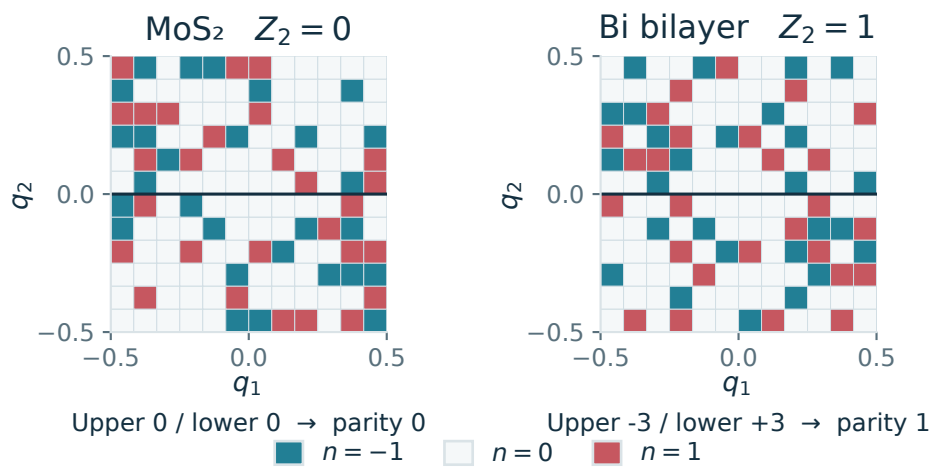
Selected pair 9-10: C_{spin} = 2

$$P\sigma_z P \phi_{\pm} = \lambda_{\pm} \phi_{\pm}, \quad C_{\pm} = \frac{1}{2\pi} \sum_p \Phi_p^{\pm}, \quad C_{\text{spin}} = \frac{C_+ - C_-}{2}$$

Bi: projected-spin sectors in pseudo-Gram geometry; full-BZ FHS flux sums.

03 Z₂ · Fukui-Hatsugai n-field

Method: Fukui-Hatsugai, JPSJ 76, 053702 (2007)



12×12 native tiles · gauge- and branch-dependent field

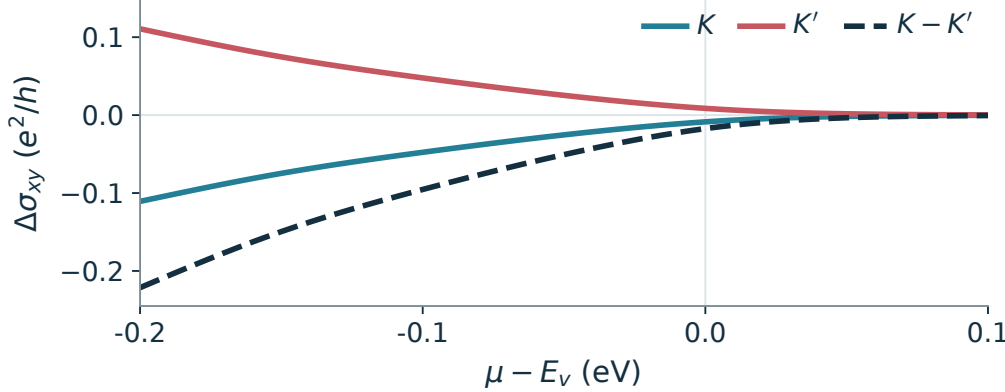
$$\nu = \left(\sum_{p \in B_{1/2}} n_p \right) \text{mod } 2$$

Fukui-Hatsugai (FH): sum the integer n-field over half the Brillouin zone.

04 Hall conductivity

Theory: Karplus-Luttinger (1954) · Sundaram-Niu (1999)

MoS₂ · regional charge Hall · 300 K



WAVECAR approximation · reference: midgap
K/K' disks: 0.35 Å⁻¹ · 36 × 36 mesh · M = 40

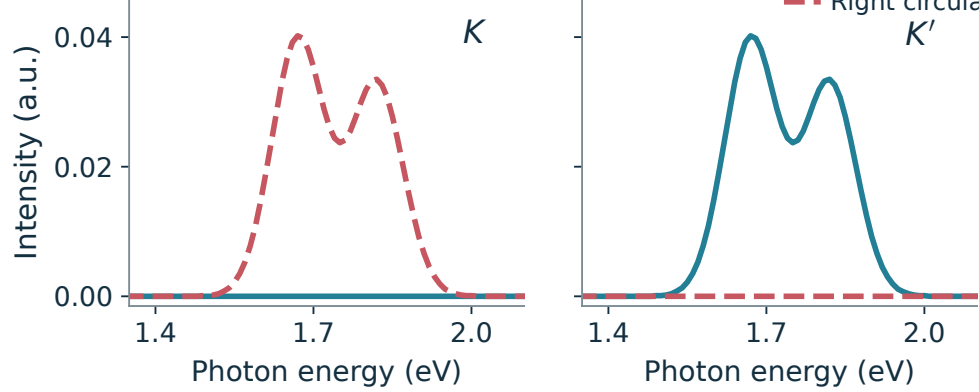
$$\Delta\sigma_{xy}^R = -\frac{e^2}{h} \sum_n \int_{R(2\pi)^2} \frac{d^2 k}{(2\pi)^2} \Delta f_n \Omega_{n,xy}$$

$\Delta f_n = f_n(\mu, T) - f_n(\mu_{\text{mid}}, T)$; R = K or K'; dashed: K - K'.

05 Circular dichroism

Optics: Yao et al., PRB 77, 235406 (2008) · Xiao et al. (2012)

MoS₂ · near-gap circular spectra



Bare-momentum spectra · Gaussian broadening 0.05 eV
k-resolved optical selection; arbitrary intensity units

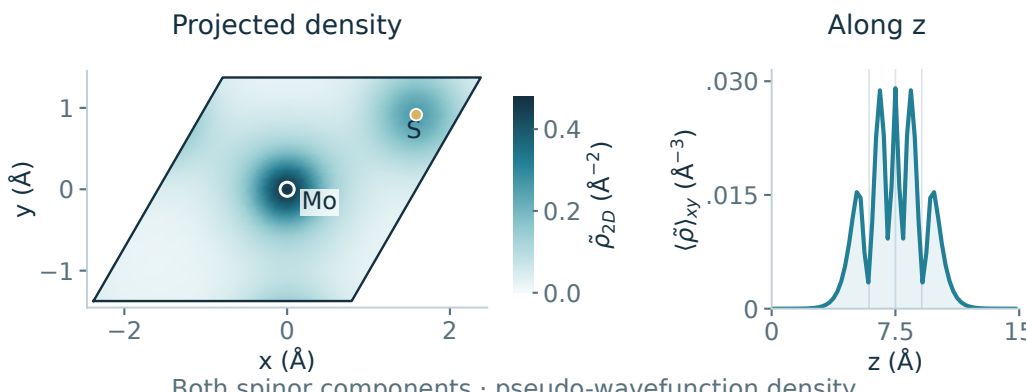
$$I_{L/R}(E_V) \propto \frac{1}{E_V^2} \sum_{v,c} |p_{cv}^x \pm i p_{cv}^y|^2 g_{\sigma}(E_V - \Delta E_{cv})$$

Left: +; right: -. g_σ: Gaussian broadening; v/c: valence/conduction.

06 Real-space wavefunction

Basis / reader: Kresse-Furthmüller (1996) · Feenstra-Widom

MoS₂ · Γ state, band 18

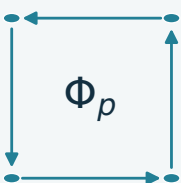


Both spinor components · pseudo-wavefunction density
No PAW augmentation · integral 0.8258, without rescaling

$$\bar{\rho}_{2D}(x, y) = \int dz \sum_s |\bar{\psi}_{\Gamma ns}(\mathbf{r})|^2$$

Both spinor components; pseudo-density without PAW augmentation.

Also available: μ/T scans · band selection · spin-sector Kubo · optional PAW spin Hall · Wannier bulk / edge spectra



Fukui-Hatsugai-Suzuki (FHS) method

Neighboring k-point overlaps → plaquette Berry flux → Chern number

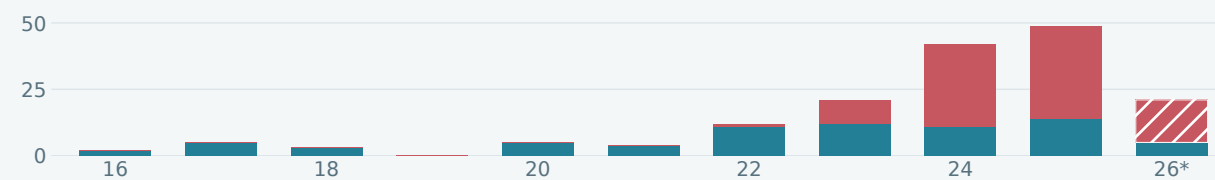
Charge / spin Chern: FHS links. Z₂: Fukui-Hatsugai (FH) n-field.

github.com/Infant83/VASPBERRY

Code, examples & technical report

Code-related PRB + PRL citations

PRB 2016 PRL 2022



164 total

OpenAlex, 4 Oct 2026; paper citations summed; *2026 YTD. GitHub: same date.

89

GitHub stars